

JOONYOUNG JEONG

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RESEARCH INTERESTS

My research interests lie in cellular network security (5G RAN and 5GC) and computer systems security. I also enjoy dissecting binaries to uncover vulnerabilities and analyze malware functionality.

EDUCATION

Iowa State University Aug 2021 – Jul 2023

M.S. in Electrical and Computer Engineering

GPA: 3.95/4.0

Relevant coursework: Communication Systems, Advanced Communications, Network Protocol Security, Cryptography, Data Analytics in ECE, Digital Signal Processing, Introduction to Deep Learning.

Korea Military Academy Mar 2009 – Mar 2013

B.E. in Mechanical Engineering

RESEARCH EXPERIENCE

Mobile Security Architecture for Defense 5G Infrastructure Jul 2026 – Dec 2026

Sponsored Research funded by ROK Army Headquarters

Analyzing defense mobile system architectures and domestic/international security policies (US DoD, NATO, 3GPP) to identify threat vectors relevant to 5G military network deployment.

Flow Matching for Missing Data Imputation [1] Mar 2026 – Jul 2026

Collaborative Research Project

Cost-Aware UE-Side Mobility Management for TN-NTN Networks [2] Sep 2025 – Aug 2026

Independent research; single-author paper accepted to IEEE MSWiM 2026

Covert authentication for securing MIMO communications [3] Jan 2022 – Jul 2023

Supervisor: Dr. Sang Kim, Iowa State University

Research Assistant, Iowa State University Aug 2022 – Jul 2023

Summarized 5G-related papers and 5GC/RAN architectures into advisor briefs

PROFESSIONAL EXPERIENCE

Rotating Faculty, Korea Military Academy Jul 2025 – Present

Teaching and advising cadets; participating in defense projects.

Planning Officer, AI R&D Center, Korea Military Academy Jul 2026 – Present

Research planning and coordination for AI-driven defense technology initiatives.

Signals Intelligence Officer, Ministry of National Defense Aug 2023 – May 2025

Malware Analysis and Cyber Threat Intelligence

Signals Intelligence Officer, Ministry of National Defense Aug 2018 – Jul 2021

COMINT

Military Intelligence Officer, Army Headquarters Mar 2013 – Aug 2018

General Military Intelligence

TEACHING

Rotating Faculty, Korea Military Academy
Introduction to Statistics (Fall 2025, Spring 2026)
Introduction to Data Mining (Fall 2025)
Linear Algebra (Spring 2026)

Jul 2025 – Present

Teaching Assistant, Iowa State University
Communication Systems II Lab (Spring 2023)
Communication Systems I Lab (Fall 2022)

Aug 2022 – Jul 2023

SKILLS

Programming Languages: Python, C/C++, x86-64 Assembly, MATLAB, Simulink, R
Analysis Tools: Burp Suite, Ghidra, IDA Pro, x64dbg, GDB, Wireshark, AFL++

HONORS & AWARDS

Chairman of the Joint Chiefs of Staff Commendation
Joint Chiefs of Staff, Republic of Korea

2024

Letter of Appreciation (SES-2)
Foreign Agency

2024

Army Merit Scholarship (Full tuition and stipend)
Ministry of National Defense

Aug 2021 – Jul 2023

PROJECTS

Intrusion Detection using Denoising Autoencoder and Deep Learning
Designed a DNN classifier using a stacked denoising autoencoder for feature learning

2023

PUBLICATIONS & WORKSHOP PAPERS

- [1] J. Jeong, S. Hwang, and Y. Kwon, “Missflow: Flow matching for uncertainty-quantified imputation of missing tabular data,” in *EIML@ICML 2026 Workshop*, Accepted, 2026.
- [2] J. Jeong, “CALM: UE-side mobility management for integrated TN-NTN networks,” in *IEEE Int. Conf. on Modeling, Analysis and Simulation of Wireless and Mobile Systems (MSWiM)*, Accepted; to appear, Paris, France, 2026.
- [3] J. Jeong, “Covert authentication for securing MIMO communications,” M.S. thesis, Iowa State University, Ames, IA, USA, 2023. [Online]. Available: <https://dr.lib.iastate.edu/bitstreams/d0ca9bbb-4436-4681-97b7-c97fa3a4ec05/download>.